

**SHRI VAISHNAV VIDYAPEETH VISHWAVIDYALAYA,
INDORE**



**INTERNSHIP REPORT
2022-23**

*A report submitted in partial fulfilment of the requirements for the Award of
Degree of*

**BACHELOR OF TECHNOLOGY
in**

Computer Science Engineering

by

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21100BTCSFBI09642

Under the Supervision of

Industry Mentor

Ms. Rinki Gupta

C.T.O.

All soft Solution private Limited

Indore

University Mentor

Prof. Jagrati Nagadiya

Assistant Professor

Computer Science & Engineering

SVIIT

(Duration: 12th January 2025 to April 2025)

Department of Computer Science Engineering
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**SHRI VAISHNAV VIDYAPEETH
VISHWAVIDYALAYA,
SHRI VAISHNAV INSTITUTE OF INFORMATION
TECHNOLOGY**



2022-23

DECLARATION

I hereby declare that work, which is being presented in the Internship Report as the partial fulfilment for the award of degree of **Bachelor of Technology in Computer Science Engineering** in the **Department of Computer Science Engineering** at **Shri Vaishnav Institute of Information Technology** of Shri Vaishnav Vidyapeeth Vishwavidyalaya Indore, is an authentic record of my work carried out under the Mentorship of **Mrs. Paridhi Nigam, Assistant Professor Department of Computer Science Engineering**. The matter embodied in this internship report has not been submitted for the award of any other degree.

Enrollment Number
21100BTCSEFI09642

Student Signature

Date:

**SHRI VAISHNAV VIDYAPEETH
VISHWAVIDYALAYA,
SHRI VAISHNAV INSTITUTE OF INFORMATION
TECHNOLOGY**



2024-2025

INTERNSHIP APPROVAL SHEET

This is to certify that **Divyanshu chouhan** Enrollment number **21100BTCSEFI09642** has successfully completed his industrial internship starting from 24th January to 30th April 2025 and has submitted the final report. His work has been found satisfactory and it is recommended to accept it as a partial fulfilment for the award of degree of **Bachelor of Technology, Computer Science Engineering** of the **Department of Computer Science Engineering** at **Shri Vaishnav Institute of Information Technology** of **Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore.**

Internal Examiner
Date:

External Examiner
Date:

INTERNSHIP COMPLETION CERTIFICATE

**SHRI VAISHNAV VIDYAPEETH
VISHWAVIDYALAYA
SHRI VAISHNAV INSTITUTE OF INFORMATION
TECHNOLOGY DEPARTMENT OF COMPUTER SCIENCE
ENGINEERING**

CERTIFICATE

This is to certify that **Divyanshu Chouhan (21100BTCSEBFI09642)** has successfully completed his industrial internship, starting from 24th January to 30th April 2025 and has submitted the final report. He has successfully completed this Internship under the Mentorship of **Prof. Jagrati Nagadiya, Assistant Professor Department of Computer Science Engineering** as a partial fulfilment of the degree of **Bachelor of Technology in Computer Science Engineering** of the **Department of Computer Science Engineering at Shri Vaishnav Institute of Information Technology of Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore** during the semester **Jan – June 2025**.

Internal Mentor
Prof. Jagrati
Nagadiya

Head & Director
Dr. Anand Rajavat

SHRI VAISHNAV VIDYAPEETH VISHWAVIDYALAYA
SHRI VAISHNAV INSTITUTE OF INFORMATION TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

ACKNOWLEDGEMENT

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Secondly, I would like to extend my sincere gratitude towards the CEO & Managing Director of **All soft Solution private Limited** for giving me an opportunity at their esteemed organization.

I also would like to thank all the people that worked along with me at Allsoft Solutions Limited. With their patience and openness, they created an enjoyable working environment. It is indeed with a great sense of pleasure and immense sense of gratitude, that I acknowledge the help of these individuals

I am highly indebted to Director and HoD **Dr. Anand Rajavat**, for the facilities and continuous support provided by them to accomplish this internship. I would also like to thank my faculty mentors **Prof. Jagrati Nagadiya** (Internal Mentor) & **Ms. Rinki Gupta** (Industry Mentor) for their constructive criticism throughout my internship. I would like to thank them for their support and advice to complete internship in above said organization. I am extremely great full to my department's staff members and friends who helped me in successful completion of this internship.

Divyanshu Chouhan
21100BTCSFBI09642
Computer Science & Engineering

ABSTRACT

The rapid advancement in technology has significantly increased the demand for secure, efficient, and user-friendly file-sharing solutions. In response to this growing need, we developed `DirectDrop`, a cutting-edge file-sharing application designed to provide users with a seamless and secure way to share files online. Built using `Flask` for the backend and `React` for the frontend, `DirectDrop` offers a wide array of features that cater to both individual and professional use cases

One of the standout features of `DirectDrop` is its ability to allow users to upload files with optional password protection, ensuring that sensitive documents remain secure during transit. Additionally, the application generates unique, shareable download links for each uploaded file, making it easy for users to distribute their files via email, messaging apps, or other communication platforms. To further simplify sharing, especially in mobile-centric environments, `DirectDrop` also includes a QR code generation feature. Users can generate a QR code for each file, allowing recipients to scan the code and access the file instantly.

Another key aspect of `DirectDrop` is its real-time upload progress tracking. Users can monitor the status of their uploads, including the percentage completed, upload speed, and estimated time remaining. This feature provides transparency and enhances the user experience by keeping users informed throughout the file transfer process. Furthermore, the application boasts a responsive and modern user interface (UI) that adapts seamlessly to both light and dark modes, ensuring accessibility and comfort for users regardless of their device settings or personal preferences.

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CHAPTER 1 – BRIEF INTRODUCTION OF THE ORGANIZATION’S BUSINESS SECTOR

Allsoft Solutions and Services Pvt Ltd, IBM Business Partner was started in 2000 having headquarters in the USA, with the establishment of three branches in India. Allsoft has been committed to providing the highest quality and needs-based services to its clients, both locally and internationally. They offer IT consulting, Industrial Training Program in various technologies, Web Development and Cloud services as well. They believe in “Making of the era with professionals” in this highly competitive workplace. Their policy towards introducing new, on demand and emerging technology makes us competitive in the market as our goals are to provide the best possible value and to lead in our markets through service and innovation. Being one of the IBM trusted and certified business partners, they offer training in IBM technologies to provide students an exposure to the IBM tools also.

ADMITTANCE

As per the emergence of the digital era, the needs of the corporate world are changing. That’s why, they swear to provide the technologies that are insanely changing our world.

IBM CERTIFIED TRAINERS

To provide the best training experience we are having our finest IBM certified trainers who will fill up brains with the knowledge of today’s growing technologies and endow for the industrial world requirements.

SMART COURSEWORK

Their coursework is carefully planned to provide the real essence of knowledge to the trainees.

They introduced an internship program that gives you a chance to enhance your objective learning and integrates your knowledge with the theory learned in the classroom. The internship program has been a well-formed channel for generating a great pipeline for the team. They are generating the perspective of professionalism in our interns as they mold them from student status to professional status by introducing them to the live projects of the cooperative world. To boost your skills and get a chance to experience a formal process within a company for the first time and start your journey with them.

DEVELOPMENT

For us leading a product from concept to implementation along with customer satisfaction is the belief.

Their solutions are not confined to products from a particular vendor or platform. But they render the services to vendors from different platforms. Based on the client's requirement, we design and deliver our products accordingly. We have strategic partnerships and domain expertise with most of the well-known names in the industry. They work on the principle that a product has to be faultless from the first stage of its

designing. They offer a complete portfolio of software solutions for different services with better customer experience.

CHAPTER 2 – OVERVIEW OF ORGANIZATION

2.1 Brief History

Allsoft Solutions and Services Pvt Ltd, IBM Business Partner was started in 2000 having headquarters in the USA, with the establishment of three branches in India. Allsoft has been committed to providing the highest quality and needs-based services to its clients, both locally and internationally. They offer IT consulting, Industrial Training Program in various technologies, Web Development and Cloud services as well. They believe in “Making of the era with professionals” in this highly competitive workplace. Their policy towards introducing new, on demand and emerging technology makes us competitive in the market as our goals are to provide the best possible value and to lead in our markets through service and innovation.

2.2 Business size

The company is a small firm, which includes 15-20 employees, having a board of directors and hardworking people to run the organization.

Allsoft Solutions, Mohali **Registration Number is MO/FEB8091/A 702.**

2.3 Product lines

Other than working on different client projects the company has collaboration to IBM by which it provides students the opportunity to get connected to IBM and explore indifferent fields like:

- Artificial Intelligence.
- Machine Learning
- IBM Cognos tool
- Full Stack Development

2.4 Competitors

- INGram
- Learn Quest

- Teach Data
- Global Knowledge

2.5 Brief summary of all the departments

The company has many departments in which it is divided into. The departments are; HR department, Front-end developing, Back-end developing, Designing, Content Writing, Customer Service, Teaching Department. Each department is unique in its own way and yet co-related and interdependent upon each other. With the cumulative effort of each department a website or app is developed and made online for usage.

CHAPTER 3: PLAN OF THE INTERNSHIP PROGRAM

3.1 Brief introduction of the branch or department:

I was appointed as a technical intern in the backend and frontend development department, where I primarily worked on developing the **DirectDrop** file-sharing application. The main task of this team was to create a secure and user-friendly platform for uploading, sharing, and downloading files. The project involved using Flask for backend development, React for frontend development, and integrating features like password protection, real-time upload progress tracking, and QR code generation. Additionally, I received practical training in technologies such as REST API's, frontend-backend integration, and responsive UI design.

3.2 The start and ending dates of internship:

The internship started on 12th January 2023, and it ended on April 2023, I have completed 14 weeks in this organization.

3.3 Name of the department in which you obtained training:

I received training in the IT and Development Cell of the organization, which focuses on creating innovative software solutions. Initially, I worked with the frontend development team to learn about React, drag-and-drop functionality, and UI/UX design principles. Subsequently, I collaborated with the backend development team to understand Flask, REST APIs, and secure file handling. In the final phase of the internship, I worked on integrating the frontend and backend components to ensure seamless functionality. Toward the end of the internship, we were divided into teams to develop and test the final version of DirectDrop.

Week 1 – Week 2	Basics of Python and Web Development
Week 3 – Week 4	Modules, Libraries, and Frameworks
Week 5 – Week 6	Backend Development and Password Protection
Week 7 – Week 8	Frontend Development and UI/UX Design
Week 9 – Week 10	Development Cycle and Testing
Week 11 – Week 15	Final Enhancements and Deployment

CHAPTER 4: TRAINING PROGRAM

4.1 Duties and Responsibilities Performed:

The internship program started on January 12th, 2023. This internship was a 15-week program which successfully ended on April, 2023. During this internship I was assigned various tasks and responsibilities.

Week 1 – Week 2 (Basics of Python and Web Development)

On the first day of the internship, there was an induction program through which I was introduced to the roadmap of this internship. Following this, I started my training with the basics of Python development, which included learning Flask for backend development and React for frontend development. I also explored the concept of secure file sharing and its importance in modern applications.

Week 3 – Week 4 (Modules, Libraries, and Frameworks)

By the third week, I collected all the necessary information and resources needed to build the DirectDrop application. I delved into the world of web development and studied various technologies such as REST APIs, drag-and-drop functionality, and QR code generation. I also learned about the Python libraries and tools we would be using, including uuid for unique filename generation and bcrypt for secure password hashing. These tools enabled us to implement features like password protection, real-time progress tracking, and file uploads.

Week 5 – Week 6 (Backend Development and Password Protection)

In the next 15 days, I learned about different types of machine learning models used in data analysis and explored their potential integration into the DirectDrop application. After acquiring the necessary skills and knowledge, I began working on implementing the core features of the application. I integrated Flask (backend) with React (frontend) to create a functional file-sharing platform that could handle file uploads, generate shareable download links, and provide optional password protection.

Week 7 – Week 8 (Frontend Development and UI/UX Design)

I started by setting up a Python environment and installing Flask and React dependencies. I learned how to use Flask to handle file uploads and serve files securely. I also gained a basic understanding of React programming, which is essential for building a responsive user interface. I delved deeper into the world of frontend development by learning about libraries like react-dropzone for drag-and-drop functionality and react-qr-code for generating QR codes. I learned how to integrate these libraries with Flask to create a seamless user experience, enabling users to upload files easily and share them via QR codes.

Week 9 – Week 10 (Development Cycle and Testing)

The team was involved in the development of DirectDrop . The team integrated Flask (backend) and React (frontend) to develop the application with features such as real-time upload progress tracking, password protection, and QR code generation. After the development was completed, the team thoroughly tested the application for accuracy and functionality. The week was focused on acquiring knowledge of how to integrate various technologies to develop a file-sharing application with various features and functionalities.

Week 11 – Week 15 (Final Enhancements and Deployment)

I completed the development cycle of DirectDrop , and it was tested for accuracy and functionality. User feedback was incorporated to improve the system, and the application was tested for scalability and accessibility. Improvements were made based on user feedback, such as optimizing performance, enhancing security, and refining the user interface. By the end of the project, I gained valuable experience in Python programming, Flask, React, and the development, testing, and deployment of a secure file-sharing application

CHAPTER 5 -LEARNING EXPERIENCE

5.1 Knowledge acquired

During my 15-week internship, I had the opportunity to learn and understand both the technical and non-technical aspects that make an organization work efficiently in the fields of web development, backend and frontend integration, secure file handling, and user interface design.

"Practice makes perfect—and this holds true for coding and development. Programming, like any other skill, demands consistent practice and sharp problem-solving abilities. Every small step, from writing individual lines of code to building complex systems, contributes to achieving success."

As a computer science student, I had prior knowledge of Python programming, web development, and basic UI/UX principles, which made it easier for me to understand and learn the new concepts introduced during my internship. Working on DirectDrop , I gained hands-on experience in building a full-stack application using Flask for the backend and React for the frontend.

I learned the importance of writing secure and efficient code, especially in the context of file-sharing applications. My mentor emphasized industry-standard coding practices, such as avoiding hardcoding sensitive information, implementing proper error handling, and ensuring code readability. These practices are crucial for maintaining the security and scalability of an application.

In the field of secure file sharing and responsive UI design, alternative solutions are essential. There is always room for improvement, even if you already know the syntax and semantics of Python programming or the solutions to problems. For instance, while working on DirectDrop , I explored multiple ways to implement password protection and upload progress tracking, which helped me refine my problem-solving skills.

Real-world insights are critical in web development and secure file sharing. You must assess the situation, understand the tools and technologies to be used, and ensure efficient data handling. My college projects helped me write maintainable and scalable code, which proved invaluable during this internship.

5.2 Skills learned

Internships are meant to be a learning experience, and while experiences and skills may vary by industry during my internship, I learned various skills and career-specific abilities. Some of those skills are listed below:

Project Management

Throughout this internship, I learned how to organize my priorities and take charge of assigned tasks. I regularly checked in with my mentor to discuss completed goals and upcoming objectives, ensuring transparency and alignment with project requirements. Managing the development cycle of DirectDrop taught me how to handle complex projects effectively.

Software Education

This internship provided an excellent opportunity to enhance my software development skills. I gained hands-on experience with Flask, React , and libraries such as react-dropzone and react-qrcode . These skills not only strengthened my resume but also made me more credible to future employers.

Teamwork

During the internship, I collaborated with team members from different departments, including frontend, backend, and database teams. I learned how to adapt my work style to fit the needs of the team and communicate effectively with colleagues. This experience helped me understand how diverse expertise contributes to the success of a project.

Frontend-Backend Integration

Working on DirectDrop, I gained practical experience in integrating the frontend and backend components. I learned how to create REST APIs using Flask and consume them in React, ensuring seamless communication between the two layers.

Learning New Tools

My work was not limited to one tool or platform. I worked with technologies such as VS Code , Postman , and Git , which were new to me. This experience taught me to step out of my comfort zone and embrace new tools to improve productivity.

Time Management

Balancing multiple tasks within tight deadlines was a key takeaway from this internship. I learned how to manage my time effectively by prioritizing tasks and maintaining a balance between work and personal life.

5.3 Observed Attitudes and Values gained

Professional communication

Being an intern, I learned how essential networking and communication are for career growth. During weekly meetings with my mentors, I demonstrated my progress and discussed challenges. This experience boosted my confidence in communicating professionally with colleagues and supervisors.

Taking criticism well

I learned to handle constructive criticism positively, which helped me grow professionally. Understanding feedback from mentors allowed me to refine my code and improve the overall quality of DirectDrop . This skill will undoubtedly benefit me in future professional environments.

Independence

During this internship, I was assigned independent tasks, such as implementing password protection and generating QR codes. These responsibilities taught me how to make decisions on my own and work with minimal guidance. Independence is a valuable trait in the professional world.

Leadership, confidence, and responsibility

As an intern, I was responsible for specific modules of DirectDrop , such as drag-and-drop functionality and real-time upload tracking. While I had direction and support from mentors, I often had to make decisions independently. This experience helped me develop leadership qualities, confidence in presenting my ideas, and a sense of responsibility toward the project's success.

Work Ethics

I developed a certain set of values based on discipline and hard work. I learned how workethics are followed in the professional world, forming good habits such as staying motivated, finishing tasks immediately, no delay, and much more help to create a great work ethic that will impress employers.

5.4 The most challenging task performed

The most challenging task performed was:

Password Protection Implementation:

One of the core features of DirectDrop is optional password protection for uploaded files. Ensuring that passwords were stored securely and verified correctly required careful planning and execution. Initially, passwords were stored in memory, which posed security risks. To address this, I researched secure methods of password storage and implemented hashing techniques to protect user data.

Real-Time Upload Progress Tracking:

Tracking upload progress in real-time was another challenging task. I had to use XMLHttpRequest 's onprogress event to monitor the percentage completion, upload speed, and estimated time remaining. Debugging and optimizing this feature required significant effort but ultimately enhanced the user experience.

CHAPTER 6: STRENGTHS, WEEKNESSES, OPPORTUNITIES, THREATS (SWOT) ANALYSIS

6.1 Strengths

User-Friendly Interface:

DirectDrop boasts a modern, responsive, and intuitive user interface that adapts seamlessly to both light and dark modes, ensuring accessibility for users across devices and preferences.

Secure File Sharing:

The application provides robust security features, including optional password protection and session management, ensuring that sensitive files remain secure during transit.

Real-Time Progress Tracking:

One of the standout strengths of DirectDrop is its ability to provide real-time upload progress tracking, including percentage completion, upload speed, and estimated time remaining. This feature enhances transparency and improves the user experience.

QR Code Generation:

The inclusion of QR code generation for download links makes it easy to share files, especially in mobile-centric environments, adding convenience and versatility to the application.

Integration of Technologies:

The seamless integration of Flask (backend) and React (frontend) ensures efficient communication between the two layers, providing a smooth and reliable user experience.

6.2 Weaknesses

Limited Scalability:

Currently, uploaded files are stored in a local directory with passwords stored in memory. This approach limits scalability and poses challenges for handling large-scale usage or persistent storage.

Lack of User Authentication:

The absence of a proper user authentication system means that anyone with access to the download link can attempt to access the file, which could be a security concern for certain use cases.

Dependency on Local Storage:

Files are stored locally on the server, which may not be suitable for long-term storage or high-volume usage. Integrating cloud storage solutions would be necessary for scalability.

Manual Password Management:

Passwords are currently stored in memory, which is insecure and inefficient. A more robust solution, such as hashing and storing passwords in a database, is needed for production-level security.

6.3 Opportunities

Cloud Storage Integration:

Integrating cloud storage services like AWS S3 or Google Cloud Storage can significantly enhance scalability and reliability, making DirectDrop suitable for enterprise-level applications.

Advanced Features:

There is potential to add advanced features such as file expiration, download statistics, user accounts, and role-based access control to cater to a wider audience and increase functionality.

Cross-Platform Expansion:

Expanding DirectDrop into a mobile app or desktop application can attract a broader user base and provide a more versatile solution for file sharing.

Monetization Potential:

By offering premium features such as enhanced security, larger file uploads, or subscription-based plans, DirectDrop can generate revenue while providing value to users.

Integration with APIs:

Integrating third-party APIs for additional functionalities, such as email notifications, analytics, or collaboration tools, can further enhance the application's capabilities.

6.4 Threats

Security Vulnerabilities:

Storing passwords in memory and relying on local storage for files can expose the application to security risks, such as unauthorized access or data breaches.

Competition from Established Platforms:

DirectDrop faces stiff competition from established file-sharing platforms like Google Drive, Dropbox, and WeTransfer, which already offer robust features and have a significant market presence.

Data Privacy Concerns:

Users are increasingly concerned about data privacy and how their files are handled. Any perceived lapses in data protection could harm the application's reputation and adoption rate.

CHAPTER 7: PROBLEM IDENTIFICATION AND SOLUTION

7.1 Problem

Problems are at the center of what developers face every day while working on software projects. Whether you are solving a problem for users (internal or external), supporting team members in overcoming challenges, or discovering new issues to address, the problems encountered can range from minor bugs to complex structural limitations.

For DirectDrop , several key challenges emerged during the development process:

In-Memory Password Storage:

Initially, passwords for password-protected files were stored in memory (file_passwords dictionary). This approach was insecure and inefficient, as it posed risks if the server restarted or faced unexpected downtime.

Lack of Scalability:

The application relied on local storage for uploaded files, which limited its ability to handle large-scale usage or persistent storage. This approach was not suitable for long-term or enterprise-level applications.

Frontend-Backend Integration Issues:

Integrating the Flask backend with the React frontend required careful handling of REST APIs. Miscommunication between the two layers occasionally led to errors, such as incorrect file uploads or failed password verifications.

Real-Time Progress Tracking Complexity:

Implementing real-time upload progress tracking using XMLHttpRequest's onprogress event was challenging. Debugging and optimizing this feature to display accurate metrics (percentage completion, upload speed, and estimated time remaining) required significant effort.

QR Code Generation Limitations:

Generating QR codes dynamically for download links worked well for smaller files but introduced performance bottlenecks when handling large files or multiple simultaneous requests.

Security Concerns:

Ensuring the security of uploaded files and protecting sensitive user data was a constant challenge. For example, storing files locally without encryption posed potential risks if unauthorized access occurred.

User Authentication Gap:

The absence of a proper user authentication system meant that anyone with access to the download link could attempt to access the file, even if password-protected.

7.2 Consequences

The above-listed problems had several consequences for the DirectDrop project:

Compromised Security:

Storing passwords in memory and relying on local storage for files increased the risk of unauthorized access or data breaches, potentially harming the application's reputation.

Limited Functionality:

The lack of scalability and cloud integration restricted the application's ability to cater to larger audiences or enterprise-level use cases.

Performance Bottlenecks:

Real-time progress tracking and QR code generation occasionally caused delays, especially when handling large files or multiple simultaneous uploads.

User Experience Gaps:

The absence of a robust user authentication system created confusion among users, as there was no clear way to manage access permissions beyond password protection.

Development Challenges:

Frontend-backend integration issues slowed down the development process and required additional debugging and testing to ensure seamless functionality. to deal with new technological projects are hindering.

7.3 Solution

To address the identified problems and their consequences, the following solutions were implemented:

Secure Password Storage:

Passwords were transitioned from in-memory storage to a secure database. Hashing algorithms like bcrypt were used to store passwords securely, ensuring they could not be easily retrieved or compromised.

Cloud Storage Integration:

Local file storage was replaced with cloud-based solutions like AWS S3 or Google Cloud Storage . This change improved scalability, reliability, and accessibility for users.

Improved Frontend-Backend Communication:

REST APIs were optimized to ensure seamless communication between the Flask backend and React frontend. Proper error handling and validation were added to prevent

miscommunication.

Optimized Real-Time Progress Tracking:

The real-time upload progress tracking feature was optimized to handle large files and multiple simultaneous uploads efficiently. Performance bottlenecks were resolved through better resource management and asynchronous processing.

Enhanced QR Code Generation:

QR code generation was optimized to handle large files and multiple requests simultaneously. Caching mechanisms were introduced to reduce redundant computations and improve performance.

User Authentication System:

A user authentication system was added to the application, allowing users to create accounts, log in, and manage their uploaded files. Role-based access control was implemented to enhance security and usability.

Formal Testing and Quality Assurance:

A formal quality assurance process was established to ensure the application met high standards of performance and security. Developers were encouraged to follow best practices, such as writing clean and maintainable code, and avoiding shortcuts during development.

Infrastructure Upgrades:

Investments were made in IT infrastructure, including servers and development tools, to create a more robust and reliable environment for building and deploying the application.

Retire Outdated Equipment:

Any outdated hardware or software used during development was replaced with modern alternatives to ensure optimal performance and compatibility. Retire equipment at proper life cycles – There must be proper management of equipment with expiring must be replaces.

CONCLUSION

This internship at Allsoft Solutions Pvt. Ltd was a valuable and enriching experience that provided me with hands-on exposure to the development of DirectDrop , a secure file-sharing application. I learned new skills and worked with cutting-edge technologies such as Flask , React , and libraries like react-dropzone and react-qr-code , gaining insights into both frontend and backend development. Working in a collaborative environment taught me how to integrate theoretical knowledge from the classroom with real-world application development.

As an individual, I faced various challenges during the internship, such as implementing password protection, ensuring secure file handling, and optimizing real-time upload progress tracking. By analyzing these problems systematically and identifying their root causes, I was able to devise effective solutions. For instance, transitioning from in-memory password storage to a secure database significantly improved the application's security. Similarly, optimizing the upload progress tracking feature enhanced the user experience and performance.

Working as part of a team, we successfully integrated the frontend and backend components of DirectDrop , ensuring seamless functionality. The experience of collaborating with developers, testers, and mentors helped me improve my problem-solving abilities and analytical thinking. Regular interactions with my mentors and teammates were instrumental in helping me identify my strengths and areas for improvement.

Upon completing the project, our team gained a thorough understanding of full-stack development, secure file handling, and responsive UI design. We developed features such as drag-and-drop file uploads, QR code generation, and real-time progress tracking, which not only enhanced the application's usability but also addressed real-world user needs. Visualizing and solving these challenges improved our creativity and ability to communicate technical concepts to both technical and non-technical stakeholders effectively.

REFERENCES AND SOURCES

Websites referred

- **www.stackoverflow.com**
A valuable resource for troubleshooting coding issues and learning best practices for Flask, React, and other technologies used in DirectDrop .
- **www.reactjs.org**
The official documentation for React, which provided guidance on building the frontend interface, including drag-and-drop functionality and responsive design.
- **www.flask.palletsprojects.com**
The official Flask documentation, which helped in setting up the backend server, handling file uploads, and implementing password protection.
- **www.freecodecamp.org**
Tutorials and articles on full-stack development, REST APIs, and integrating Flask with React.
- **www.geeksforgeeks.org**
Articles on Python programming, Flask, and secure file handling techniques.
- **www.github.com**
Explored open-source projects for inspiration and best practices in file-sharing applications.

Books referred

"Flask Web Development" by Miguel Grinberg

"Learning React" by Alex Banks and Eve Porcello

"Python Crash Course" by Eric Matthes

"Full Stack Development with Flask and React" by Sagar Shivani

APPENDICES



Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore
Shri Vaishnav Institute of Information Technology

Notification: SVVV/N-083/2019

Date: February 4, 2025

Fortnightly Report of the Student/Intern – Form 1

Progress Report no.: 1

Enrollment No:- 21100BTCSEFBI09642		Name:- Divyanshu Chouhan	
	Period	Date From	Date To
Name of the Industry :- Allsoft Solutions Pvt. Ltd External Mentor Name :- Ms. Rinki Gupta Email Id :- rinki.ibmce@gmail.com Mobile no. :- +91 8054474039	15 Days	24/01/2025	04/02/2025
Main points of the fortnight	Week 1		
	• Setting Up Environment For Project		
	• Project and Tech Stack is finalized		
	• Analysis of similar apps, Pizza Share		
	• Installation of web RTC on local server		
	• Research on api and linking		
	Week 2		
	• Finalize frontend languages		
	• Basic UI Structure Using Tailwind CSS		
	• Set up WebRTC peer connections and create basic signaling logic.		
	• Implement file selection and transfer logic on the frontend.		
	• Designed and built UI for file selection and transfer.		
	• Add error handling for potential issues (e.g., failed file transfer, connection loss).		

Intern Signature

External Mentor Signature

Internal Mentor Signature



Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore
Shri Vaishnav Institute of Information Technology

Notification: SVVV/N-083/2019

Date: February 19, 2025

Fortnightly Report of the Student/Intern – Form 2

Progress Report no.: 2

Enrollment No:- 21100BTCSEFI09642		Name:- Divyanshu Chouhan	
	Period	Date From	Date To
Name of the Industry :- Allsoft Solutions Pvt. Ltd External Mentor Name :- Ms. Rinki Gupta Email Id :- rinki.ibmce@gmail.com Mobile no. :- +91 8054474039	15 Days	05/01/2025	19/02/2025
Main points of the fortnight	Week 3		
	• Set up WebRTC peer connection logic.		
	• Integrate WebRTC with UI		
	• Encrypt file data before transmission.		
	• Report critical issues before the next phase.		
	• Password protection to files		
	• Resolving bugs and issues		
	Week 4		
	• Add animations for file transfer progress.		
	• Implement guest & authenticated user modes.		
	• Conduct load testing for concurrent transfers.		
	• Document code changes and update README.		
	• Fix UI/UX bugs reported by QA.		
• Summarize testing results and prepare a final report.			

Intern Signature

External Mentor Signature

Internal Mentor Signature



Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore
Shri Vaishnav Institute of Information Technology

Notification: SVVV/N-083/2019

Date: March 06, 2025

Fortnightly Report of the Student/Intern – Form 3

Progress Report no.: 3

Enrollment No:- 21100BTCSEFI09642		Name:- Divyanshu Chouhan	
	Period	Date From	Date To
Name of the Industry :- Allsoft Solutions Pvt. Ltd External Mentor Name :- Ms. Rinki Gupta Email Id :- rinki.ibmce@gmail.com Mobile no. :- +91 8054474039	15 Days	19/02/2025	06/03/2025
Main points of the fortnight	Week 5		
	• Implement pause/resume functionality for file transfers.		
	• Improve the file transfer progress UI with detailed statistics.		
	• Test and optimize the app for mobile browsers.		
	• Review each other's code and optimize.		
	• Fix reported issues and optimize performance.		
	• Resolving bugs and issues		
	Week 4		
	• Implement end-to-end encryption for file transfers.		
	• Set up a beta release environment for user testing.		
	• Write API and WebRTC integration documentation.		
	• Conduct a final team review and prepare for deployment.		
	• Fix UI/UX bugs		
	• Report to tester		

Intern Signature

External Mentor Signature

Internal Mentor Signature



Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore
Shri Vaishnav Institute of Information Technology

Notification: SVVV/N-083/2019

Date: March 21, 2025

Fortnightly Report of the Student/Intern – Form 4

Progress Report no.: 4

Enrollment No:- 21100BTCSEFI09642		Name:- Divyanshu Chouhan	
	Period	Date From	Date To
Name of the Industry :- Allsoft Solutions Pvt. Ltd External Mentor Name :- Ms. Rinki Gupta Email Id :- rinki.ibmce@gmail.com Mobile no. :- +91 8054474039	15 Days	06/03/2025	21/03/2025
Main points of the fortnight	Week 7		
	• Implement bandwidth management for better speed control.		
	• Implement session expiration to prevent unauthorized access.		
	• Compile performance reports and identify bottlenecks.		
	• Improve error messages and tooltips for better user guidance.		
	• Fix reported issues and optimize performance.		
	• Review and clean up code for better maintainability.		
	Week 8		
	• Improve reconnection logic for interrupted transfers.		
	• Improve error handling for failed or incomplete transfers.		
	• Handle very large file transfers smoothly.		
	• Explore integration options for future cloud storage support.		
	• Fix UI/UX bugs		
• Discuss progress, challenges, and next steps.			

Intern Signature

External Mentor Signature

Internal Mentor Signature



Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore
Shri Vaishnav Institute of Information Technology

Notification: SVVV/N-083/2019

Date: April 04, 2025

Fortnightly Report of the Student/Intern – Form 5

Progress Report no.: 5

Enrollment No:- 21100BTCSEFI09642		Name:- Divyanshu Chouhan	
	Period	Date From	Date To
Name of the Industry :- Allsoft Solutions Pvt. Ltd External Mentor Name :- Ms. Rinki Gupta Email Id :- rinki.ibmce@gmail.com Mobile no. :- +91 8054474039	15 Days	21/03/2025	04/04/2025
Main points of the fortnight	Week 9		
	• Set up MongoDB for storing file logs.		
	• Created schemas for users and file sessions.		
	• Developed REST APIs for login/signup.		
	• Integrated JWT for session management.		
	• Implemented API for file metadata.		
	• Handled secure API routing.		
	Week 10		
	• Logged peer sessions in DB.		
	• Optimized DB queries.		
	• Conducted load testing on backend.		
	• Fixed API response bugs.		
	• Created Swagger docs for API.		
• Configured environment variables for deployment.			

Intern Signature

External Mentor Signature

Internal Mentor Signature